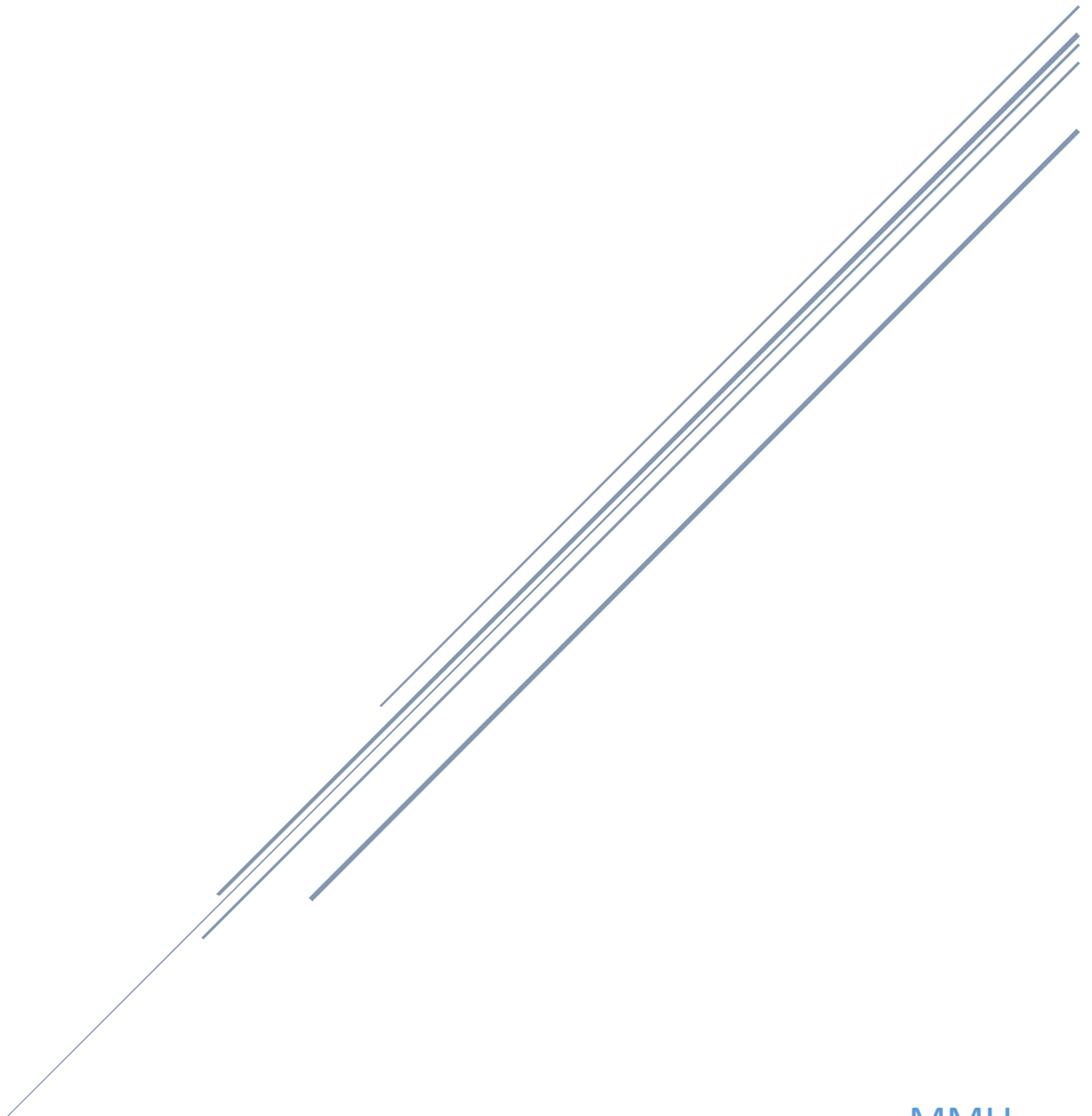


SJF ASSIGNMENT

TO BE SUBMITTED TO MADAM MANGO



MMU
BY 24/05/2026

Group Assignment: Preemptive Shortest Job First (SJF / SRTF) CPU Scheduling Algorithm

1. Assignment Overview

In this group assignment, you will work in teams of **five (5)** to design and implement the **Preemptive Shortest Job First (SJF)** CPU scheduling algorithm—also known as **Shortest Remaining Time First (SRTF)**—using a **programming language of your choice** (Python, Java, C, or C++ are recommended).

Your program will simulate how an operating system schedules processes using preemptive SJF and must compute key CPU scheduling performance metrics.

2. Learning Objectives

By completing this assignment, you should be able to:

- Understand CPU process scheduling and the mechanics of preemptive SJF.
- Implement process scheduling algorithms programmatically.
- Generate and interpret a **Gantt Chart** of process execution.
- Calculate CPU scheduling metrics:
 - Waiting Time
 - Turnaround Time
 - Completion Time
- Work collaboratively in a structured team environment.
- Produce a technical report and deliver a team presentation.

3. Task Requirements

Your group must design and implement a program that:

A. Accepts the following inputs:

1. Number of processes
2. For each process:
 - Arrival time
 - CPU Burst time

B. Performs CPU scheduling using:**Preemptive Shortest Job First (SRTF)**

At every unit of CPU time, the process with the **shortest remaining burst time** must be selected.

If a newly arrived process has a shorter remaining time, it must **preempt** the current process.

C. Produces the following outputs:

1. **Gantt Chart** showing the execution sequence.
2. **Completion Time (CT)** for each process.
3. **Turnaround Time (TAT)** for each process.
4. **Waiting Time (WT)** for each process.
5. **A well-formatted output table.**

4. Deliverables

Each group must submit:

1. Source Code

- Your full program (Python, Java, C, or C++).
- Proper comments and clean formatting.
- Should run without errors.

2. Report (5–10 pages)

Your report must include:

- **Title Page** with group members and registration numbers
- **Introduction to CPU Scheduling & SJF**
- **Algorithm Explanation (SRTF)**
- **Flowchart or Diagram**
- **Screenshot(s) of your running program**
- **Gantt Chart**
- **Results table with metrics**
- **Discussion of challenges and lessons learned**
- **Conclusion**

3. Presentation (5–10 minutes)

Your group must prepare a short oral presentation covering:

- Your approach
- Code overview
- Gantt chart & results
- Challenges and reflections

5. Group Work Requirements

- The group consists of **exactly 5 members**.
- Every member must contribute to at least one area:
 - Coding
 - Debugging
 - Report writing
 - Diagram creation
 - Presentation
- Include a **group contribution declaration** in your report.

6. Academic Integrity

- All work must be original.
- Plagiarism or copying code from the internet will result in zero marks.
- You may use online resources for learning but must write your own program.

7. Deadline

The deadline for submission of this assignment is 22nd/05/2026

Late submissions may attract penalties.